

NEARBY GOVERNMENT SERVICES FINDER

PROJECT REPORT

TEAM MEMBERS:

1. NANDAM BALA SAI SURYA (25A31A0549)
2. SATHI SIVA CHARAN REDDY (25A31A0560)
3. LAKKOJU PAVAN BALA MURALI (25A31A0540)
4. SHEIK HAMDUNISHA (25A31A0526)
5. MALLAVARAPU SRI PAVITRA DEVI (25A31A0516)

GUIDED BY:

MR. Y. MANAS KUMAR

ASSISTANT PROFESSOR, CSE DEPARTMENT

PRAGATI ENGINEERING COLLEGE (A), SURAMPALEM

1. PROJECT DESCRIPTION

The” **Nearby Government Services Finder**” is a console-based C++ application designed to help users quickly locate essential government services available near their area. This project provides a simple and user-friendly menu-driven interface through which users can search for important public services such as hospitals, police stations, fire stations, post offices, municipal offices, government schools, and RTO offices.

The system stores predefined service details such as service category, name, area, phone number, and distance, and displays relevant information based on the user’s selected category.

Purpose: The main purpose of this project is to help users easily identify and access nearby government services in a simple and organized way. In real life, people often need quick information about essential public service centers, especially during emergencies or for day-to-day needs. This application simulates such a system in a C++ console environment.

This project is especially useful for students learning C++ programming concepts, academic mini-project demonstrations, understanding menu-driven applications, practicing structures, arrays, and searching techniques, and building a real-world service locator concept in a simple format.

How It Works: The application uses a predefined list of government services. Each service contains the following details: Service Category, Service Name, Area / Location, Phone Number, and Distance from the user (sample data). The user interacts with the system through a menu and selects a service category such as Hospitals, Police Stations, Fire Stations, Post Offices, Municipal Offices, Government Schools, or RTO Offices. After the selection, the program searches the stored data and displays all matching services in that category.

Key Features:

- Simple menu-driven interface
- Search government services by category
- Displays service details clearly
- Uses Kakinada-based sample data for realism
- Easy to understand and implement
- Demons

trades real-world use of C++ concepts

- Suitable for academic mini-project submission

2. SYSTEM ARCHITECTURE

1. SYSTEM OVERVIEW: The Nearby Government Services Finder is designed as a menu-driven C++ console application that helps users search and display nearby government services using predefined local data.

The system works based on the following concepts:

- User input through menu selection
- Searching service records by category
- Displaying matched government service details
- Repeating the process until the user exits

2. CORE COMPONENTS

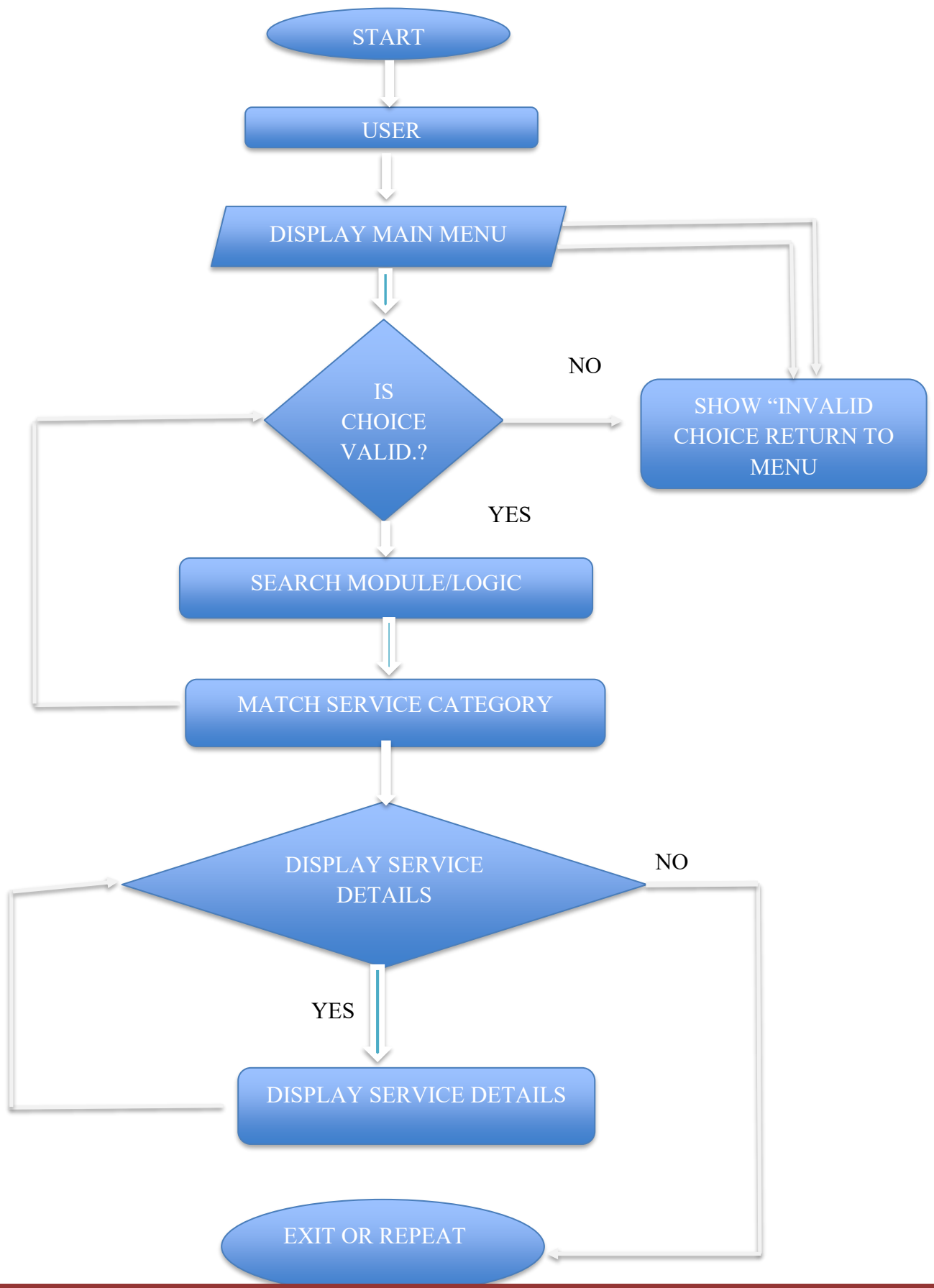
2.1 Data Input Layer: This layer contains the predefined data used by the program. It stores service records such as Category, Service Name, Area / Location, Contact Number, and Distance. Example service categories include Hospital, Police, Fire, Post, Municipal, School, and RTO. The data is stored using Structure (struct) and Array of structures.

2.2 User Interaction Layer: This layer handles the interaction between the user and the program. It includes displaying the main menu, accepting the user's choice, allowing category selection, and showing output based on the selected option.

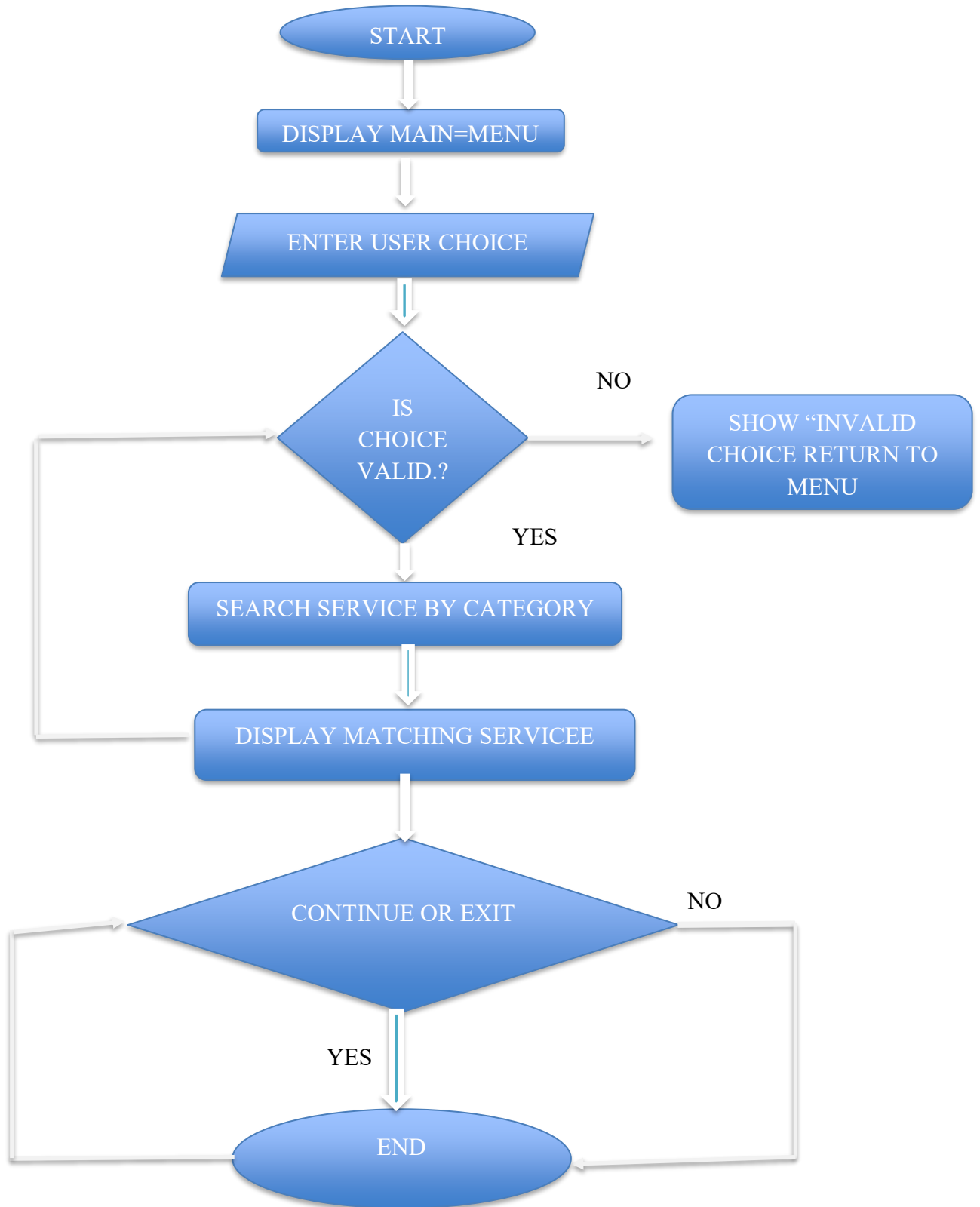
2.3 Processing Layer: This is the main logic layer of the application. It performs reading the user's selected menu option, matching the selected category with stored service data, searching through the array of services, and displaying matching service records.

2.4 Output Layer: This layer displays the result to the user. It shows Service Name, Area / Location, Contact Number, and Distance. It can also display all services, specific category services, invalid choice messages, and exit messages.

System Architecture Flow:



FLOW CHART: NEAR GOVERNMENT SERVICE FINDER



4. ALGORITHM

STEP 1: Start the program

STEP 2: Define a structure to store government service details

STEP 3: Store predefined service data in an array

STEP 4: Display the main menu

STEP 5: Accept the user's choice

STEP 6: Compare the selected option with available service categories

STEP 7: Search matching services in the array

STEP 8: Display service details if found

STEP 9: If user selects "Show All", display all service records

STEP 10: If user selects "Exit", terminate the program

STEP 11: Stop

5. MODULES USED

1. Menu Module: Displays all service category options and accepts user input.
2. Data Storage Module: Stores service details in an array of structures and maintains predefined Kakinada-based service data.
3. Search Module: Searches services based on the selected category and matches the category with stored records.
4. Display Module: Shows service details in a formatted way and displays all services or filtered results.
5. Exit Module: Safely terminates the program when the user selects exit.

6. TECHNOLOGIES USED

- Programming Language: C++
- Concepts Used:
 - Structures
 - Arrays
 - Functions
 - Loops
 - Conditional Statements
 - Searching Logic
- Platform: Console-based Application
- Compiler Used: Dev-C++ / G++ / Code::Blocks

7. ADVANTAGES

- Simple and easy to use
- Beginner-friendly C++ project
- Helps locate important public services quickly
- Can be extended for real-time applications
- Demonstrates practical use of programming concepts
- Uses local area-based data for realism

8. FUTURE ENHANCEMENTS

- Add search by area name
- Add sorting by nearest distance
- Store data in external files
- Add admin option to add/update/delete services
- Integrate with real-time GPS and maps
- Develop GUI version using C++ frameworks like Qt
- Create a web/mobile version in future

9. CONCLUSION

The Nearby Government Services Finder is a simple and useful C++ mini project that demonstrates how programming can be applied to solve a practical real-world problem. It helps users search and view nearby government services such as hospitals, police stations, post offices, and municipal offices through a menu-driven console interface. This project successfully uses core C++ concepts like structures, arrays, functions, loops, and searching techniques. It is an effective academic mini project that can be further improved into a real-time public service locator system.